

D3 Proposed Package Insert

REFRAT Mycophenolate Mofetil Capsules USP 250mg

REFRAT Mycophenolate Mofetil Tablets USP 500mg

COMPOSITION:

Each capsule contains Mycophenolate Mofetil 250mg

Each film coated tablet contains Mycophenolate Mofetil 500mg

PRODUCT DESCRIPTION:

250mg: White to off white blend of Mycophenolate Mofetil filled in size 1 hard gelatin capsules with ivory Cap and ivory body printed "SAL" on cap & "726" on body in black.

500mg: Pinkish brown coloured, capsule shaped, film coated tablets with "SAL" engraved on one side and engraved "725" on other side.

PHARMACOLOGY:

Mycophenolate mofetil, the 2-morpholinoethyl ester of mycophenolic acid (MPA), is an immunosuppressive agent. Mycophenolate mofetil is a prodrug that has little pharmacologic activity until hydrolyzed in vivo to mycophenolic acid (MPA), the pharmacologically active metabolite. MPA is a potent, selective, uncompetitive and reversible inhibitor of inosine monophosphate dehydrogenase (IMPDH), and therefore inhibits the de novo pathway of guanosine nucleotide synthesis without incorporation in to DNA. MPA has more potent cytostatic effects on lymphocytes than on other cells.

PHARMACOKINETICS:

Absorption

Mycophenolate mofetil is well absorbed following oral administration, having an absolute bioavailability of approximately 94% (based on mycophenolic acid and. undergoes complete presystemic metabolism to the active metabolite, MPA. Mycophenolate mofetil can be measured systemically during intravenous infusion; however, after oral administration it is below the limit of quantitation (0.4 µg/mL).

Food had no effect on the extent of absorption (MPA AUC) of mycophenolate mofetil administered at doses of 1.5 g twice daily to renal transplant patients. However, MPA C_{max} was decreased by 40% in the presence of food.

Distribution

Secondary increases in plasma MPA concentration are usually observed at approximately 6-12 hours post-dose, consistent with enterohepatic recirculation. At clinically relevant concentrations, MPA is 97% bound to plasma albumin.

Metabolism

MPA is conjugated primarily with glucuronyl transferase to form the phenolic glucuronide of MPA (MPAG), which is not pharmacologically active. In vivo, MPAG is converted to free MPA via enterohepatic recirculation.

Elimination

93% of an orally administered dose of mycophenolate mofetil is excreted in urine (about 87% mainly as the phenolic glucuronide of MPA and negligible amount of less than 1% of dose is excreted as MPA) and 6% is excreted in faeces. At clinically encountered concentrations MPA and MPAG are not removed by hemodialysis. However, at high MPAG concentrations ($>100 \mu\text{g/mL}$), small amounts of MPAG are removed. By interfering with enterohepatic circulation of the drug, bile acid sequestrants, such as cholestyramine, reduce MPA AUC.

INDICATIONS:

REFRAT is indicated for the prophylaxis of organ rejection in patients receiving allogeneic renal, cardiac and hepatic transplants. It should be used concomitantly with cyclosporine and corticosteroids.

REFRAT is indicated for induction and maintenance treatment of lupus nephritis. It should be used concomitantly with corticosteroids.

DOSAGE AND ADMINISTRATION

The initial dose of REFRAT should be given as soon as the patients can tolerate oral medication following renal, cardiac or hepatic transplantation.

Recommended dosage:

For prophylaxis of renal rejection: A dose of 1 g administered twice a day (daily dose of 2 g) for use in renal transplant patients.

For prophylaxis of cardiac rejection: A dose of 1.5 g administered twice a day (daily dose of 3 g) for use in cardiac transplant patients.

For prophylaxis of hepatic rejection: A dose of 1.5 g administered twice a day (daily dose of 3 g) for use in hepatic transplant patients.

Induction and maintenance treatment of lupus nephritis: A dose of 1g administered twice a day (daily dose of 2g).

Special dosage instructions

Patients with neutropenia

If neutropenia develops (absolute neutrophil count $<1.3 \times 10^3/\mu\text{L}$), dosing with mycophenolate mofetil should be interrupted or the dose reduced.

Patients with severe renal impairment

In renal transplant patients with severe chronic renal impairment (glomerular filtration rate $<25 \text{ mL/min/1.73m}^2$), outside of the immediate post-transplant period or after treatment of acute or

refractory rejection, doses greater than 1 g administered twice a day should be avoided. No data are available for cardiac or hepatic transplant patients with severe chronic renal impairment. No data are available for cardiac transplant patients with severe chronic impairment.

Patients with delayed renal graft function post-transplant

No dose adjustments are needed in patients experiencing delayed renal graft function post-operatively.

Patients with severe hepatic impairment

No dose adjustments are needed for renal patients with severe hepatic parenchymal disease. No data are available for cardiac transplant patients with severe hepatic parenchymal disease.

No data are available for cardiac transplant patients with severe hepatic parenchymal disease.

Elderly (≥65 years)

The recommended oral dosage of 1 g twice daily for renal transplant patients and 1.5 g twice daily for cardiac transplant patients are appropriate for elderly patients.

Children

Safety and efficacy in paediatric patients have not been established. Very limited pharmacokinetic data are available for paediatric renal transplant patients. No data are available for paediatric patients receiving cardiac transplants.

CONTRAINDICATIONS

REFRAT tablet/capsule is contraindicated in patients with hypersensitivity to mycophenolate mofetil, mycophenolic acid or any of its excipients.

REFRAT is contraindicated during pregnancy due to its mutagenic and teratogenic potential (see Use in Special Populations: Pregnancy).

REFRAT is contraindicated in women of childbearing potential not using highly effective contraceptive methods (see Use in Special Populations: Pregnancy).

REFRAT is contraindicated in women who are breastfeeding (see Use in Special Populations: Breastfeeding).

WARNING AND PRECAUTIONS

Immunosuppressed patients are at increased risk for opportunistic infections, including activation of latent viral infections. These include BK virus associated nephropathy which has been observed in patients receiving immunosuppressants. These infections may lead to serious, including fatal outcomes.

As in all patients receiving immunosuppressive regimens involving combinations of drugs, patients receiving Mycophenolate mofetil as part of an immunosuppressive regimen are at increased risk of developing lymphomas and other malignancies, particularly of the skin. The risk appears to be related to the intensity and duration of immunosuppression rather than to the use of any specific agent. As with all patients at an increased risk for skin cancer, exposure to sunlight and UV light should be limited by wearing protective clothing and using a sun screen with a high protection factor.

Patients receiving Mycophenolate mofetil should be instructed to report immediately any evidence of infection, unexpected bruising, bleeding or any other manifestation of bone marrow depression.

Patients should be advised that during treatment with Mycophenolate mofetil, vaccinations may be less effective and the use of live attenuated vaccines should be avoided.

Because Mycophenolate mofetil has been associated with an increased incidence of digestive system, adverse events, including infrequent cases of gastrointestinal tract ulceration, hemorrhage, and perforation. Mycophenolate mofetil should be administered with caution in patients with active digestive system disease.

Because Mycophenolate mofetil is an inosine monophosphate dehydrogenase (IMPDH) inhibitor, on theoretical grounds it should be avoided in patients with rare hereditary deficiency of hypoxanthine-guanine phosphoribosyl-transferase (HGPRT) such as Lesch-Nyhan and Kelley-Seeg-miller syndrome.

It is recommended that Mycophenolate mofetil should not be administered concomitantly with azathioprine because both have the potential to cause bone marrow suppression and such concomitant administered has not been studied.

In view of the significant reduction in the AUC of MPA by cholestyramine, caution should be used in the concomitant administration of Mycophenolate mofetil with drugs that interfere with enterohepatic recirculation because of the potential to reduce the efficacy of Mycophenolate mofetil.

Administration of doses greater than 1 g twice daily to renal patients with severe chronic renal impairment should be avoided. No dose adjustment is recommended for post-transplant patients with delayed renal graft function, but patients should be carefully monitored. No data are available for cardiac or hepatic transplant patients with severe renal impairment.

Elderly patients may be at increased risk of adverse events compared with younger individuals.

Laboratory monitoring

Patients on mycophenolate mofetil should have complete blood counts weekly during the first month of treatment, twice monthly for the second and third months, then monthly through the first year. In particular, patients receiving mycophenolate mofetil should be monitored for neutropenia. The development of neutropenia may be related to Mycophenolate mofetil, concomitant medications, viral infection or some combination of these causes. If neutropenia develops (absolute neutrophil count $<1.3 \times 10^3/\mu\text{L}$), dosing with Mycophenolate mofetil should be interrupted or the dose reduced and the patient should be carefully observed.

DRUG INTERACTIONS

Acyclovir: Higher MPAG and acyclovir plasma concentrations were observed when mycophenolate mofetil was administered with acyclovir than when the drugs administered alone. Because MPAG plasma concentrations are increased in the presence renal impairment, as are acyclovir concentrations, the potential exists for mycophenolate and acyclovir, to compete for tubular secretion, further increasing the concentrations of both drugs.

Antacids with magnesium and aluminium hydroxides: Absorption of mycophenolate mofetil was decreased when the drug was administered with antacids.

Cholestyramine: Following single dose administration of 1.5 g of mycophenolate mofetil to normal healthy subjects pre-treated with 4 g t.i.d. of cholestyramine for 4 days, there was a 40% reduction in the AUC of MPA. Caution should be used during concomitant administration or with drugs that interfere with enterohepatic circulation.

Cyclosporin A: Cyclosporin A pharmacokinetics were unaffected by mycophenolate mofetil.

Ganciclovir: Based on the results of a single dose administration study of recommended doses of oral mycophenolate and i.v. ganciclovir and the known effects of renal impairment on the pharmacokinetics of mycophenolate mofetil and ganciclovir, it is anticipated that coadministration of these agents (which compete for mechanisms of renal tubular secretion) will result in increases in MPAG and ganciclovir concentration. No substantial alteration of MPA pharmacokinetics is anticipated and a mycophenolate mofetil dose adjustment is not required. In patients with renal impairment in which mycophenolate mofetil and ganciclovir, are coadministered, the patient should be monitored carefully.

Oral contraceptives: The pharmacokinetics of oral contraceptives were unaffected by coadministration of mycophenolate mofetil. Study of coadministration of mycophenolate mofetil and combined oral contraceptives showed no clinically relevant influence of mycophenolate mofetil on serum levels of progesterone, LH and FSH, thus indicating no influence of mycophenolate mofetil on the ovulation-suppressing action of the oral contraceptives.

Trimethoprim/sulphamethoxazole: No effect on the bioavailability of MPA was observed.

Tacrolimus: In renal transplant patients: Stable renal transplant patients receiving cyclosporin and mycophenolate mofetil (1 g twice daily) showed about a 30% increase in MPA plasma AUC and about a 20% decrease in MPAG plasma AUC when cyclosporin was replaced with tacrolimus. MPA C_{max} was not affected, while MPAG C_{max} was reduced by approximately 20%. The mechanism of this finding is not well understood. Increased biliary secretion of MPAG accompanied with increased enterohepatic recirculation of MPA may be partly responsible for the finding, since the elevation of MPA concentrations associated with tacrolimus administration was more pronounced in later portions of the concentration - time profile (4-12 hours after dosing). For patients on tacrolimus, the dose of mycophenolate mofetil should not exceed 1 g twice a day. Patients should be carefully observed and managed appropriately. In another study in renal transplant patients, it was shown that the tacrolimus concentration did not appear to be altered by mycophenolate mofetil.

Other interactions: Coadministration of Probenecid with mycophenolate mofetil in monkeys raises the plasma AUC of MPAG 3- fold. Thus, other drugs known to undergo renal tubular secretion may compete with MPAG and thereby raise plasma concentrations of MPAG or the other drug undergoing tubular secretion.

Live vaccines: Live vaccines should not be given to patients with an impaired immune response. The antibody response to other vaccines may be diminished.

USE IN PREGNANCY AND LACTATION

Pregnancy

REFRAT is contraindicated during pregnancy and in women of childbearing potential not using highly effective contraceptive methods. (see Contraindications).

Before the start of treatment, female and male patients of reproductive potential must be made aware of the increased risk of pregnancy loss and congenital malformations and must be counseled regarding pregnancy prevention, and planning.

Prior to starting therapy with REFRAT, female patients of childbearing potential must have two negative serum or urine pregnancy tests with a sensitivity of at least 25 mIU/mL; The second test should be performed 8-10 days after the first one and immediately before starting REFRAT. Repeat pregnancy tests should be performed during routine follow-up visits. Results of all pregnancy tests should be discussed with the patient. Patients should be instructed to consult their physician immediately should they become pregnant.

Due to the mutagenic and teratogenic potential of mycophenolate, women of child bearing potential should use two reliable forms of contraception simultaneously, including at least one highly effective method, before beginning mycophenolate therapy, during therapy, and for six weeks following discontinuation of therapy, unless abstinence is the chosen method of contraception.

Sexually active men are recommended to use condoms during treatment and for at least 90 days after cessation of treatment. Condom use applies for both reproductively competent and vasectomised men, because the risks associated with the transfer of seminal fluid also apply to men who have had a vasectomy. In addition, female partners of male patients are recommended to use highly effective during treatment and for total of 90 days after the last dose of REFRAT.

Congenital malformations, including multiple malformations have been reported post-marketing in children of patients exposed to mycophenolate in combination with other immunosuppressants during pregnancy. The following malformations were most frequently reported:

- Facial malformations such as cleft lip, cleft palate, micrognathia and hypertelorism of the orbits;
- Abnormalities of the ear (e.g. abnormally formed or absent external/middle ear) and eye (e.g. coloboma, microphthalmos);
- Malformations of the fingers (e.g. polydactyly, syndactyly, brachydactyly);
- Cardiac abnormalities such as atrial and ventricular septal defects;
- Oesophageal malformations (e.g. oesophageal atresia);
- Nervous system malformations (such as spina bifida).

In the medical literature, malformations in children from mycophenolate-exposed pregnancies have been reported in 23% to 27% of live births. For comparison, the risk of malformations is estimated at approximately 2% of live births in the overall population and at approximately 4% to 5% in solid organ transplant patients treated with immunosuppressants other than mycophenolate.

Cases of spontaneous abortions have also been reported in patients exposed to mycophenolate, mainly in the first trimester. In the medical literature, the risk has been reported at 45% to 49% following mycophenolate exposure, compared to a reported rate between 12 and 33% in solid organ transplant patients treated with other immunosuppressants.

Studies in animals have shown reproductive toxicity.

Breastfeeding

Studies in rats have shown mycophenolate mofetil to be excreted in milk. It is not known whether mycophenolate mofetil is excreted in human milk. Due to the potential for serious adverse reactions in nursing infants from mycophenolate mofetil, a decision should be made whether to discontinue nursing or to discontinue the drug, taking into account the importance of the drug to the mother.

REFRAT is contraindicated during breastfeeding due to the potential for serious adverse reactions in nursing infants (see Contraindications).

Studies in rats have shown mycophenolate to be excreted in milk. It is not known whether this medicine is excreted in human milk.

ADVERSE REACTIONS

ADVERSE DRUG REACTIONS

General disorders and administration site conditions: (uncommon) de novo purine synthesis inhibitors-associated acute inflammatory syndrome

General disorders and administration site conditions

De novo purine synthesis inhibitors-associated acute inflammatory syndrome has been described from post-marketing experience as a paradoxical proinflammatory reaction associated with mycophenolate mofetil and mycophenolic acid, characterised by fever, arthralgia, arthritis, muscle pain and elevated inflammatory markers. Literature case reports showed rapid improvement following discontinuation of the medicinal product.

The adverse event profile associated with the use of immunosuppressive drugs is often difficult to establish owing to the presence of underlying diseases and the concurrent use of many other medications.

The most frequently reported adverse effects associated with mycophenolate mofetil therapy are diarrhoea, leukopenia, sepsis, vomiting, higher frequency of infections, including opportunistic infections (e.g., CMV infections, herpes zoster, herpes simplex, candida infections, aspergillosis, and *Pneumocystis carinii pneumonia*).

Adverse reactions occurring in 20% or more of patients receiving mycophenolate mofetil include pain (e.g. abdominal, chest, back), fever, headache, anaemia (e.g., hypochromic), thrombocytopenia, leukocytosis, urinary tract infection, abnormal renal function, hypertension, hypotension, cardiovascular disorder, tachycardia, oedema (e.g., peripheral), hypercholesteremia, hypokalemia, hyperkalemia, hyperglycemia, increases in blood urea nitrogen (BUN) and serum creatinine concentration, increased lactic dehydrogenase, hypomagnesemia, hypocalcemia, constipation, dyspepsia, nausea, vomiting, anorexia, abnormal liver function test results, cough, dyspnoea, lung disorder, sinusitis, pleural effusion, rash, tremor, insomnia, dizziness, anxiety, and paresthesia.

Other adverse effects:

Gastrointestinal: colitis (sometimes caused by cytomegalovirus), pancreatitis, isolated cases of intestinal villous atrophy.

Disorders of immunosuppression: serious life-threatening infections such as meningitis and infectious endocarditis have been reported occasionally; there is evidence of a higher frequency of certain types of infections such as tuberculosis and atypical mycobacterial infection.

Post-marketing experience:**Congenital Disorders**

Congenital malformations have been reported post-marketing in children of patients exposed to mycophenolate in combination with other immunosuppressants during pregnancy (see Use in Pregnancy).

Pregnancy, Puerperium and Perinatal Conditions

Cases of spontaneous abortions mainly in the first trimester in patients exposed to mycophenolate have been reported (see Use in Pregnancy).

OVERDOSAGE AND TREATMENT

The experience with overdose of mycophenolate mofetil in humans is very limited. MPA cannot be removed by haemodialysis. However, at high MPAG plasma concentrations ($>100\text{ }\mu\text{g/mL}$), small amounts of MPAG are removed. Bile acid sequestrants, such as cholestyramine, can remove MPA by increasing excretion of the drug.

EFFECT ON ABILITY TO DRIVE AND USE MACHINES

REFRAT may have a moderate influence on the ability to drive and use machines.

Patients should be advised to use caution when driving or using machines if they experience adverse drug reactions such as somnolence, confusion, dizziness, tremor or hypotension during treatment with REFRAT (see section Undesirable Effects).

SPECIAL REMARKS***Handling and disposal:***

Because mycophenolate mofetil has demonstrated teratogenic effect in rats and rabbits (see 'Use in pregnancy and lactation'), REFRAT tablets should not be crushed and REFRAT capsules should not be opened or crushed. Avoid inhalation or direct contact with skin or mucous membranes of the powder contained in REFRAT capsules. If such contact occurs, wash thoroughly with soap and water, rinse eyes with plain water.

STORAGE

Store below 30°C. Protect from light.

AVAILABILITY

250mg Capsules: Blister pack of 5x10's and 10x10's.

500mg Tablets: Blister pack of 5x10's.

For further information, please consult your physician or pharmacist.

MANUFACTURED BY

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